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Thesis for obtaining the academic degree
Bachelor of Science

Probing Fast Radio Burst Environments: Cross-Correlating Deep and Shallow FRB Surveys with KiDS DR5

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Abstract

The abstract is a short summary of the thesis in English. Together with the German summary, both summaries have to fit on this page.

Kurzfassung

Hier steht eine Kurzfassung der Arbeit in deutscher Sprache inklusive der Zusammenfassung der Ergebnisse. Zusammen mit der englischen Zusammenfassung muss sie auf diese Seite passen.

Contents

1 Introduction

- relevance
- emerging probes –> different papers cite
- whats plan: frb, galaxy survey –> combination of both

2 Theoretical Background

2.1 Cosmological Background

The foundation of modern cosmology is based on the *Cosmological Principle*, which states that the Universe is homogeneous and isotropic on large scales. Homogeneity implies that the large-scale properties of the Universe are statistically uniform across different locations, whereas isotropy means that no direction is preferred. Although this principle is introduced as a fundamental assumption in modern cosmology, it also represents a hypothesis that can be tested through observations. Current astronomical evidence strongly supports the validity of the cosmological principle on the largest observable scales [26, 12].

Based on the cosmological principle, the Universe can be described by the *Friedmann-Lemaître-Robertson-Walker* (FLRW) metric

$$ds^2 = -c^2 dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

with the spatial comoving coordinates (r, θ, ϕ), the speed of light c, and the cosmic time t [3]. The parameter k describes the curvature of the Universe, which can take values of -1, 0, or $+1$ for open, flat, and closed geometries, respectively. The scale factor $a(t)$ describes the relative expansion of the Universe and is conventionally normalized to its present value,

$$a(t_0) = 1.$$

For earlier cosmic times, the scale factor satisfies $a(t) < 1$, whereas in an expanding Universe it increases with time and can exceed unity in the future. A static Universe would correspond to $\dot{a}(t) = 0$, resulting in a vanishing Hubble parameter [10].

The expansion rate of the Universe is quantified by the Hubble parameter,

$$H(t) = \frac{\dot{a}(t)}{a(t)},$$

which describes the relative change of the scale factor with cosmic time. At the present time

$$H_0 = H(t_0)$$

is referred to as the Hubble constant [3].

The expansion of space introduces a distinction between physical distances and comoving distances. While the physical distance between two objects changes with time due to the expansion of the Universe, the comoving distance remains constant for objects that follow the general expansion of space. The physical or so called proper distance is related to the comoving distance χ by

$$D(t) = a(t)\chi.$$

Thus, comoving coordinates remove the effect of the cosmic expansion and provide a convenient description of the large-scale structure of the Universe [11].

The expansion of the Universe can be observed through the cosmological redshift. Due to the expansion of space, the wavelength of photons emitted by distant objects is stretched during their propagation. The redshift is defined as

$$z = \frac{\lambda - \lambda_0}{\lambda_0},$$

where λ_0 is the emitted wavelength and λ is the observed wavelength. The redshift is directly related to the scale factor through

$$1 + z = \frac{a(t_0)}{a(t_e)} \tag{2.1}$$

with t_e denoting the time of emission. Consequently, the observed redshift provides information about the expansion history of the Universe and can be used to determine cosmological distances. The line-of-sight comoving distance to an object at redshift z is given by

$$\chi(z) = \chi_H \int_0^z \frac{\mathrm{d}z'}{E(a(z'))},$$

where $\chi_H = c/H_0$ is the Hubble distance and $E(a) = H(a)/H_0$ is the dimensionless Hubble parameter [9].

2.2 Friedmann Equations and Cosmic Expansion

Following the cosmological principle, the matter and energy content of the Universe can be described by a perfect-fluid energy-momentum tensor. In combination with the FLRW metric, Einstein's field equations reduce to the *Friedmann equations*

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda}{3}$$

which determine the evolution of the scale factor $a(t)$. Here ρ and p denote the energy density and pressure of the cosmic fluid, Λ is the cosmological constant, and G is the gravitational constant [10].

The critical density at present time is defined as the total energy density for which the Universe is spatially flat ($k = 0$) and is given by,

$$\rho_c = \frac{3H_0^2}{8\pi G}.$$

The ratio between the density of a cosmological component and the critical density defines the corresponding density parameter,

$$\Omega_{\mathrm{i}} = \frac{\rho_i}{\rho_c}.$$

with Ω_m, Ω_r, Ω_Λ, and Ω_k denoting the density parameters of matter, radiation, vacuum (dark) energy, and curvature, respectively.

Consequently, the Friedmann equation can be expressed in terms of the density parameters [1]:

$$H^2 = H_0^2\left[\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda\right].$$

Using Equation 2.1, the dimensionless Hubble parameter can also be written as

$$E(z) = \sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_{k,0}(1+z)^2 + \Omega_{\Lambda,0}}.$$

2.2.1 The ΛCDM Model

The current standard model of cosmology is the so-called ΛCDM model. It describes the evolution of the Universe assuming the presence of cold dark matter (CDM) and a cosmological constant Λ, which is associated with dark energy and the accelerated expansion of the Universe. The model is in good agreement with a wide range of cosmological observations [18]. For the late-time Universe (low redshift) probed by extragalactic sources, the radiation density is negligible and the expansion history can be approximated by the contributions from matter, curvature, and dark energy. Assuming a spatially flat Universe, the curvature term also vanishes, yielding [7]

$$E(z) \approx \sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}}.$$

2.3 Fast Radio Bursts

Fast Radio Bursts (FRB) are bright, transient radio pulses with durations of milliseconds or less and peak flux densities ranging from 50 mJy to 100 Jy [20]. They have been detected across a frequency range of 110 MHz [22] to 8 GHz [31].

The isotropic-equivalent energy distribution of the observed FRB population follows a power-law distribution described by

$$N(E_{\text{iso}}) \propto E_{\text{iso}}^{-1.8},$$

indicating that high-energy bursts are increasingly rare compared to lower-energy events. The negative power-law index of approximately -1.8 describes the decrease in the occurrence rate with increasing isotropic-equivalent energy.

Furthermore, the implied isotropic energies of FRBs are within a few orders of magnitude of those released in gamma-ray bursts and supernova explosions, suggesting that FRBs may originate from highly energetic astrophysical processes [20, 30].

Consequently, a broad range of progenitor models has been proposed, including neutron stars, magnetars, white dwarfs, black holes ultra-luminous X-ray sources, active galactic nuclei, and more exotic scenarios such as superconducting cosmic strings [21].

Despite releasing within a single millisecond an amount of energy comparable to what the Sun radiates over three days [20], FRBs are typically located at cosmological distances. As their radio emission propagates across the expanding Universe, it spreads over an immense area, so that the signal received on Earth is roughly a thousand times weaker than that of a mobile phone located on Earth's Moon [27] [ZU POPULAERWISSENSCHAFTLICH ???].

The signal from FRBs is dispersed as it propagates through the ionized interstellar and intergalactic medium, causing lower-frequency components to arrive later than higher-frequency components. This makes FRBs appear as a frequency-dependent sweep in time, which can be described by the following relation:

$$DM = \frac{2\pi m_e c}{e^2} \Delta t \left(\frac{1}{\nu_{\text{low}}^2} - \frac{1}{\nu_{\text{high}}^2} \right)^{-1},$$

where ν_{low} and ν_{high} are the lower and higher observing frequencies, Δt is the time delay between the arrival of the two frequency components, c is the speed of light, e is the elementary charge, and m_e is the electron mass. The dispersion measure DM quantifies the integrated column density of free electrons between the source and the observer:

$$DM \propto \int_0^d n_e(l)\, dl,$$

where n_e represents the free electron density and d the distance between the source and the observer [20].

The origin of FRBs is established through their large DM, which exceed the expected contribution from the Galactic interstellar medium along the corresponding line of sight. For a source located within the Milky Way, typical values predicted by Galactic electron-density models are of the order of $DM_{\text{MW}} \lesssim 50\,\text{pc cm}^{-3}$.

In contrast, observed FRBs often exhibit significantly larger values $DM_{\text{FRB}} \gtrsim 1000\,\text{pc cm}^{-3}$ [23].

The first FRB, later named the Lorimer Burst, was identified in archival pulsar survey data acquired in 2001 and published in 2007 [14].

Initially, only a small number of FRBs had been detected, leading to the assumption that they represented rare, singular events. The situation changed decisively with the commissioning of CHIME in 2018, whose wide field of view drove an explosion in detections and increased the known FRB population by several orders of magnitude. The latest CHIME/FRB catalog contains 4539 detected burst events originating from 3641 unique FRB sources, including 981 bursts from 83 known repeating sources [5]. As shown in Figure 2.1, the known FRB sources exhibit an approximately isotropic distribution across the sky, consistent with an extragalactic population and suggesting no preferred spatial direction. The shown distribution is based on the FRB sample available up to 2022 and serves as an illustration of the large-scale spatial distribution of FRBs.

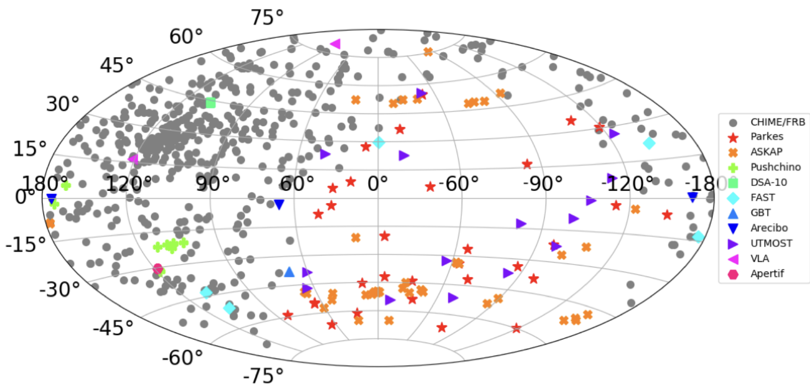

Figure 2.1: Aitoff projection of the known FRB population in Galactic coordinates based on observations available up to 2022. The sample is predominantly composed of CHIME/FRB detections, whose non-uniform sensitivity and sky coverage influence the observed distribution. After correcting for these observational biases, the underlying FRB population is consistent with an isotropic distribution expected for a cosmological source population [21].

2.3.1 Repeating FRBs

While the majority of FRBs have been observed as one-off events, a subset of sources has been found to emit multiple bursts over time. The existence of so-called repeaters immediately places a fundamental constraint on emission physics: any cataclysmic, single-event progenitor mechanism is directly ruled out for these sources, since the central engine must be capable of sustaining burst activity over timescales of years [20].

Some repeating FRBs also exhibit periodic modulation of their activity windows, as observed in FRB 20180916B [17], suggesting either a binary system geometry or precession of the emitting object. Whether all FRBs are intrinsically capable of repeating remains an open question [21].

2.3.2 Progenitor Mechanisms

A broad classification of proposed FRB progenitor models distinguishes between two main categories: catastrophic, one-off events and non-catastrophic scenarios involving long-lived central engines. Catastrophic models can account only for non-repeating FRBs and are therefore incompatible with the observed behavior of repeating sources. In contrast, non-catastrophic models allow for multiple burst episodes over extended periods. Among the proposed progenitors, compact stellar remnants are considered the most promising candidates. In particular, neutron stars and magnetars are strongly supported by both theoretical models and an increasing body of observational evidence [20].

Magnetars are neutron stars distinguished by surface magnetic fields many orders of magnitude stronger than those of ordinary radio pulsars, making them the most strongly magnetised stellar remnants known [20]. They arise primarily through the core collapse of massive stars in supernova explosions and are therefore intrinsically young objects whose formation rate is expected to closely mirror the cosmic star formation history (SFH), preferentially found in or near regions of active star formation [19]. The landmark detection in April 2020 of an FRB-like burst from the Galactic magnetar SGR 1935+2154 provided the most direct empirical link between magnetars and FRBs [4]. Population synthesis work using the CHIME/FRB catalogue finds that the hypothesis of FRBs tracking the SFH is not ruled out by current data, with a small delay preferred, consistent with a scenario in which young magnetars formed through core-collapse supernovae serve as the dominant progenitor class [16].

Neutron stars formed through delayed channels represent a second important progenitor class, motivated by FRBs localised to environments inconsistent with active star formation. FRBs 180924 and 190523, found in massive, low-star-formation-rate galaxies with large offsets from the host centre, display environments similar to those of short gamma-ray burst hosts associated with mergers of binary neutron-star or neutron star-black hole binaries [28]. The globular cluster localisation of FRB 20200120E reinforces this point, as such an old environment is is generally not associated with recent star formation. Isolated core-collapse supernovae occur within approximately 3 Myr to 48 Myr after a burst of star formation, whereas binary interactions can delay some core-collapse events to about 200 Myr [29]. Accretion-induced collapse (AIC) and compact-object merger channels exhibit much broader delay-time distributions extending to several gigayears [25]. As a consequence, this progenitor population does not closely track the cosmic SFH but evolves more slowly with redshift [13].

These two progenitor classes predict observably distinct redshift distributions [13], forming the theoretical basis for the host-population bias modelling adopted in this work, in which each class is assigned a redshift-dependent bias b(z) encoding how strongly its occurrence rate evolves relative to the cosmic star-formation rate. The parameterisation and specific values are detailed in chapter 3.

WAS IST MIT DEN FRAGEN AN KLARA WEGEN DER EDGE CASES SIEHE SLACK?????

2.4 Galaxy Surveys and Catalogs

- Galaxy Survey DR5 stage 3 –> stage 54 KiDS, how measured
- COSMOS2020 Catalog ?
- Comparison of the Data Sets

2.5 Structure Formation

Within the ΛCDM framework, the large-scale structure of the Universe emerged from the gravitational amplification of small primordial density perturbations. According to the gravitational instability paradigm, quantum fluctuations seeded during an inflationary phase were stretched to cosmological scales and subsequently grew into the filaments, voids, and galaxy clusters observed today [2, 24]. The growth of perturbations depends sensitively on the cosmic expansion history, and the overall amplitude of matter fluctuations at the present time is conventionally parameterised by σ_8 [1].

2.6 Correlation Functions and Angular Power Spectrum

The statistical properties of the large-scale matter distribution are quantified through correlation functions and their Fourier-space counterparts, the power spectra.

The density contrast $\delta(\mathbf{x})$ is defined as the fractional deviation of the local matter density $\rho(\mathbf{x})$ from the average cosmic density $\bar{\rho}$:

$$\delta(\mathbf{x}) := \frac{\rho(\mathbf{x}) - \bar{\rho}}{\bar{\rho}}.$$

The two-point correlation function of the density contrast is then defined as

$$\xi(\mathbf{y}) := \langle \delta(\mathbf{x})\, \delta(\mathbf{x}+\mathbf{y}) \rangle,$$

where the angle brackets denote an ensemble average over all positions $\mathbf{x}$. It describes the excess probability of finding two tracers at a given separation $|\mathbf{y}|$ relative to a random distribution. Its Fourier transform yields the three-dimensional power spectrum $P(k)$, defined by

$$\langle \hat{\delta}(\mathbf{k})\, \hat{\delta}(\mathbf{k}') \rangle =: (2\pi)^3 P(k)\, \delta_{\mathrm{D}}(\mathbf{k}-\mathbf{k}'),$$

with the Dirac delta function δ_{D} ensuring statistical homogeneity, and where $P(k)$ depends only on the magnitude $k = |\mathbf{k}|$, reflecting statistical isotropy [1].

HIER NOCHMAL UEBERLGEN BZW NACHFRAGEN WIESO HOMOGEN USW...

When the tracers of interest are observed as projected distributions on the celestial sphere, the appropriate framework shifts from three-dimensional Fourier analysis to a spherical harmonic decomposition. Since the sky is curved, a flat-sky Fourier transform is insufficient on large angular scales; instead, any field $f(\theta,\phi)$ on the sphere can be expanded as

$$f(\theta,\phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m}\, Y_{\ell m}(\theta,\phi),$$

where $Y_{\ell m}$ are the spherical harmonics and the harmonic coefficients are given by

$$a_{\ell m} = \int f(\theta,\phi)\, Y^*_{\ell m}(\theta,\phi)\, \mathrm{d}\Omega.$$

The multipole moment ℓ plays the role of an angular wavenumber, with small ℓ corresponding to large angular scales and large ℓ to small-scale structure.

The angular auto-power spectrum C_ℓ is defined as the variance of the harmonic coefficients per multipole,

$$\langle a_{\ell m}\, a^*_{\ell' m'} \rangle =: \langle C_\ell \rangle\ \delta_{\ell\ell'}\, \delta_{mm'},$$

where different multipoles are statistically independent for an isotropic field. The angular power spectrum can be estimated as

$$C_\ell = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2,$$

given a set of observed harmonic coefficients [8].

The cross-correlation between two distinct projected fields A and B is captured by the angular cross-power spectrum

$$\langle a^A_{\ell m}\, a^{B*}_{\ell' m'} \rangle =: \langle C_\ell^{AB} \rangle\ \delta_{\ell\ell'}\, \delta_{mm'}.$$

BIS HIER ALLES GECHECKT AUSSER HALT DIE FETTGEDRUCKTEN SACHEN OBEN A non-zero detection of C_ℓ^{AB} indicates that the two populations spatially overlap on angular scale $\sim 180°/\ell$. The cross-power spectrum can be related to the underlying three-dimensional matter power spectrum $P(k, z)$ through a line-of-sight projection. Under the Limber approximation, valid for $\ell \gtrsim \mathcal{O}(10)$, this takes the compact form [15]

$$C_\ell^{AB} = \int \frac{\mathrm{d}z}{c} \frac{H(z)}{\chi^2(z)}\, W^A(z)\, W^B(z)\, P\!\left(k = \frac{\ell + 1/2}{\chi(z)},\, z\right),$$

where $W^A(z)$ and $W^B(z)$ are the window functions encoding the radial selection of each tracer population, $\chi(z)$ is the comoving distance, and $H(z)$ is the Hubble parameter. For $A = B$, this expression reduces to the angular auto-power spectrum. The cross-correlation signal thus depends on the redshift overlap of the two populations, their individual bias parameters with respect to the matter field, and the underlying matter power spectrum. This makes the FRB–galaxy cross-power spectrum a direct probe of the spatial distribution and host-population properties of FRBs, forming the central observable of this work. Bei correlation functions —> power spectrum und Fourier function !!!!

- Two-Point Correlation Function
- Auto-Correlation
- Cross-Correlation

3 Methodology

Irgendwo das hier: Since luminous tracers such as galaxies do not perfectly follow the underlying dark matter distribution, their spatial statistics are related to the matter field through a bias parameter b, a concept that is formalised in section 2.6 and forms a central ingredient of this work [6].

3.1 Redshift Distribution

3.1.1 Shallow vs. Deep

3.2 Background Correlation

3.3 Auto-Correlation Analysis

3.4 Cross-Correlation Analysis

4 Results

4.1 Fisher Matrix

5 Conclusion & Outlook

6 Structure of the thesis

A possible structure for the thesis is as follows:

1. **Introduction**
   The *short* introduction should present the motivation for the work and provide an overview of the upcoming chapters.
2. **Theoretical Background**
   Keep theoretical background concise; refer to textbooks for standard material.
3. **Results**
   The main part of the thesis where the conducted work and findings are presented.
4. **Summary and Outlook**
   Summary of findings, possible improvements, and future directions.

Use chapters for the top-level structure. Avoid chapters with only one section.

Diese Vorlage ist auf die Kompilierung mit `lualatex` ausgelegt. Als Dokumentenklasse wird die KOMA-ScriptKlasse `scrbook` verwendet. Falls Sie Änderungen am Layout vornehmen möchten, lesen Sie die KOMA-Script-Dokumentation:

Eine umfangreiche Einführung in die moderne Verwendung von LaTeX gibt es hier: , lesenswert ist außerdem das LaTeX-Tabu

Um dieses Dokument vollständig zu erstellen sind maximal vier Programmläufe nötig:

1. `lualatex BachelorArbeit.tex`
2. `biber BachelorArbeit.bcf`
3. `lualatex BachelorArbeit.tex`
4. `lualatex BachelorArbeit.tex`

Beim ersten Lauf des LaTeX-Compilers werden die Kapitel, Links und zitierten Bibliographieeinträge in Hilfsdateien geschrieben.

Dann ist ein Lauf des Programms `biber` nötig, welches die benötigten Einträge aus der Hilfsdatei einliest, die Einträge aus der `.bib` Datei einliest, sortiert und formatiert und in eine weitere Hilfsdatei schreibt.

Beim nächsten LaTeX-Lauf werden dann diese Hilfsdateien eingelesen und Literatur- und Inhaltsverzeichnis erstellt.

Manchmal ist ein vierter Lauf nötig, falls sich durch das einfügen des Literaturverzeichnisses Seitenzahlen verändert haben.

Das Tool `latexmk` übernimmt dies mit nur einem Programmaufruf und führ nur so viele Aufrufe durch, wie nötig sind.

```
latexmk --lualatex BachelorArbeit.tex
```

Eine gute Option ist es, den LaTeX Output in einem anderen Ordner zu erzeugen, dies ist mit der `--output-directory` Option möglich:

```
latexmk --output-directory=build --lualatex BachelorArbeit.tex
```

6.1 Erstellen des Ausgabedokuments mit Make

Für diese Vorlage wird ein Makefile zur Verfügung gestellt, welches automatisch alle Schritte ausführt, die für das fertige Dokument nötig sind. Die Ausgabe erfolgt dabei in den Unterordner `build/`. Make prüft, ob die Quelldateien verändert wurden, falls nicht, werden auch keine Befehle ausgeführt.

Falls Sie das Makefile benutzen möchten, sollten Sie alle Abhängigkeiten eintragen (Eigene Dateien für Kapitel, Plots, etc.).

Download und weitere Informationen zu Make gibt es unter. Die Befehle sind für die Bash ausgelegt. Wenn Sie sie unter Windows nutzen wollen, benötigen Sie einen Bash-Emulator, wie Git Bash, Download unter möglich. Wenn Sie Make installiert haben, rufen Sie einfach in der Konsole im Verzeichnis der Arbeit den Befehl `make`.

6.2 Erstellen des Ausgabedokuments mit Texmaker

6.2.1 Einrichten der nötigen Befehle

Ein beliebter Editor für alle Betriebssysteme ist Texmaker, Download unter . Damit Texmaker das Dokument korrekt kompiliert, fügen sie einen benutzerdefinierten Befehl hinzu:

1. Klicken sie oben in der Menüleiste auf *Benutzer/in*
2. Klick auf *Eigene Befehle*

3. Klich auf *Eigene Befehle editieren*, dort können Sie bis zu 5 eigene Befehle definieren
4. Geben Sie dem Befehl unter *Menüeintrag* einen Namen und tragen sie folgende Befehle in das Befehlsfeld ein:
`latexmk --lualatex --interaction=batchmode --halt-on-error %.tex |`
5. Bestätigen Sie mit *OK*

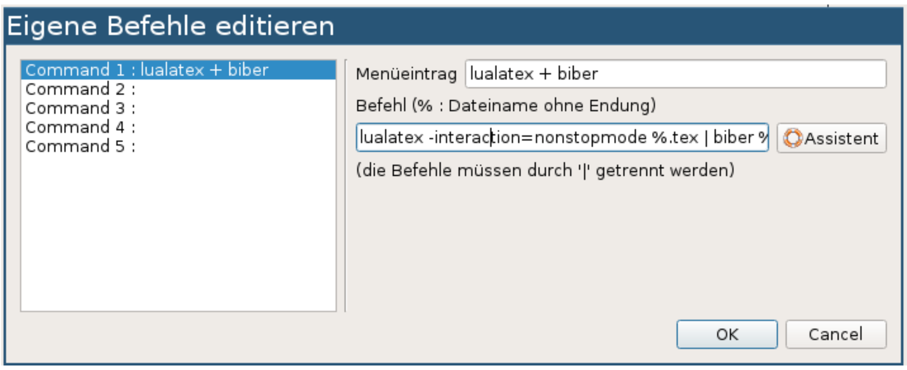

Figure 6.1: Screenshot zur Erstellung des Kompilier-Befehls in Texmaker

In Abbildung 6.1 ist ein Screenshot des Befehlsmenü gezeigt. Ihren Befehl können Sie nun im Drop-Down-Menü zum Kompilieren des Dokuments auswählen und mit einem Klick auf den Pfeil starten.

6.2.2 Aufräumen

Nach einem LaTeX-Fehler ist es oft notwendig, die erstellten Hilfsdateien zu löschen. Klicken Sie hierzu auf *Werkzeuge→Aufräumen*.

Bitte beachten Sie beim Schreiben der Arbeit folgende Konventionen bzw. Grundlagen:

- **Abschnitte und Zeilenumbrüche**
  Es sollten im Fließtext keine Zeilenumbrüche mit \\ erzwungen werden. Schreiben Sie höchsten einen Satz in eine Code-Zeile. Absätze werden im Code mit einer Leerzeile markiert und dann entsprechend der Einstellung von `parskip` in der Dokumentenklasse gesetzt.

- **Kursiv/Aufrecht**

  - Variablen und physikalische Größen werden kursiv gesetzt.

- Einheiten werden immer aufrecht und mit einem halben Leerzeichen Abstand zur Zahl gesetzt. Nutzen Sie `siunitx`!
- Mathematische Konstanten und Funktionen werden ebenfalls aufrecht gesetzt. Zum Beispiel die Eulersche Zahl e, das imaginäre i und das infinitesimale d. Im Mathematikmodus können Sie dies mit dem Befehl `\mathrm{}` erreichen. Für die Funktionen stellt LaTeX Befehle bereit, z.B. `\arccos`.
- Integrand und ein dx sollten ebenfalls durch ein kleines Leerzeichen (`\,`) getrennt werden.

6.3 Zahlen und Einheiten

Jede Zahl, jede Einheit und jede Zahl mit Einheit sollte mit Hilfe der in dem Paket `siunitx` zur Verfügung gestellten Befehle gesetzt werden. Grundsätzlich gilt: Einheiten werden aufrecht gesetzt und haben ein kleines Leerzeichen (`\,`) Abstand zu ihrer Zahl. Werden Fließkommazahlen ohne `siunitx` gesetzt, entsteht ein hässlicher Leerraum zwischen Komma und erster Nachkommastelle, da LaTeX das Komma nicht als Dezimaltrennzeichen, sondern als Satzzeichen interpretiert.

Das Paket wurde mit deutschen Spracheinstellungen (also mit Komma als Dezimaltrennzeichen und · zwischen Zahl und Zehnerpotenz) geladen, sowie mit den Einstellungen, dass die Standardabweichung stets durch ± abgetrennt wird und Einheiten falls nötig als Brüche ausgegeben werden.

Table 6.1: Beispiele für siunitx

Befehl	Ergebnis
`\num{1.2345}`	1,2345
`\num{1.2e3}`	$1{,}2 \cdot 10^3$
`\num{1.2 +- 0.2}`	$1{,}2 \pm 0{,}2$
`\num{10000}`	10 000
`\si{\meter\per\second}`	m/s
`\SI{1.2(1)}{\micro\ampere}`	$(1{,}2 \pm 0{,}1)$ µA
`\SI{1.2\pm0.1e3}{\kilo\gram\per\cubic\meter}`	$(1{,}2 \pm 0{,}1) \cdot 10^3\,\mathrm{kg/m^3}$

Das Paket stellt unter anderem die drei wichtigen Befehle

- `\num{Zahl}`,
- `\si{Einheit}` und

- `\SI{Zahl}{Einheit}`

zur Verfügung. Diese Befehle sollten stets genutzt werden, wenn Zahlen angegeben werden. Sie funktionieren sowohl im Text- als auch im Mathematikmodus. In Tabelle 6.1 sind einige Beispiele aufgetragen. Bitte lesen Sie die Dokumentation .

6.4 Das Literaturverzeichnis

Im Text können Sie mit `\cite{kürzel}` zitieren. Seitenzahlen geben Sie in eckigen Klammern an: `\cite[10]{kürzel}`.

6.5 Figures

When preparing plots, use sufficiently large axis labels, bold enough font weights and distinguishable colors. Ideally, plots should use the same font and font size as the document. Create plots at the correct size and include them with the optional argument `scale=1` when possible.

Prefer vector graphics (PDF) and use raster images (PNG, JPG) only when necessary. End figure captions with a period.

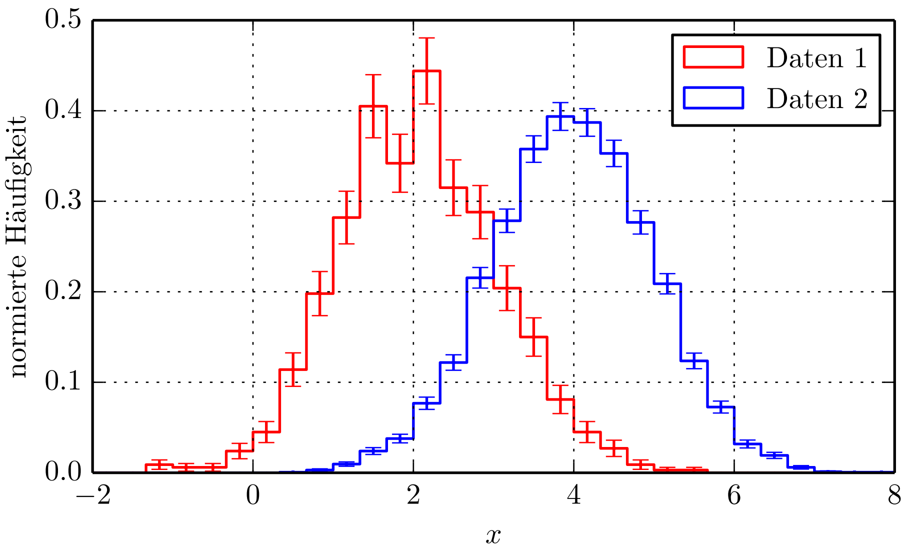

Figure 6.2: An example histogram with error bars for two datasets; font size and style match the document.

6.6 Tables

Keep tables as simple as possible and avoid excessive rules. In most cases three horizontal rules are sufficient: above, below and between the header and the body. The `booktabs` package provides `\\toprule`, `\\midrule` and `\\bottomrule` for this purpose. The `siunitx` package supplies the `S` column type for nicely aligned numbers.

An example is Table 6.2. ngerman

Table 6.2: Example table using the `S` column type from `siunitx`.

p / Pa	T / K
1024,23	273,15
1025,31	274,5
1026,27	276,2

A Appendix

This appendix can contain supplementary material such as code listings, drawings or other extended material that does not fit into the main text. In most cases, all relevant results should be presented in the main part of the thesis and an appendix is optional.

Bibliography

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